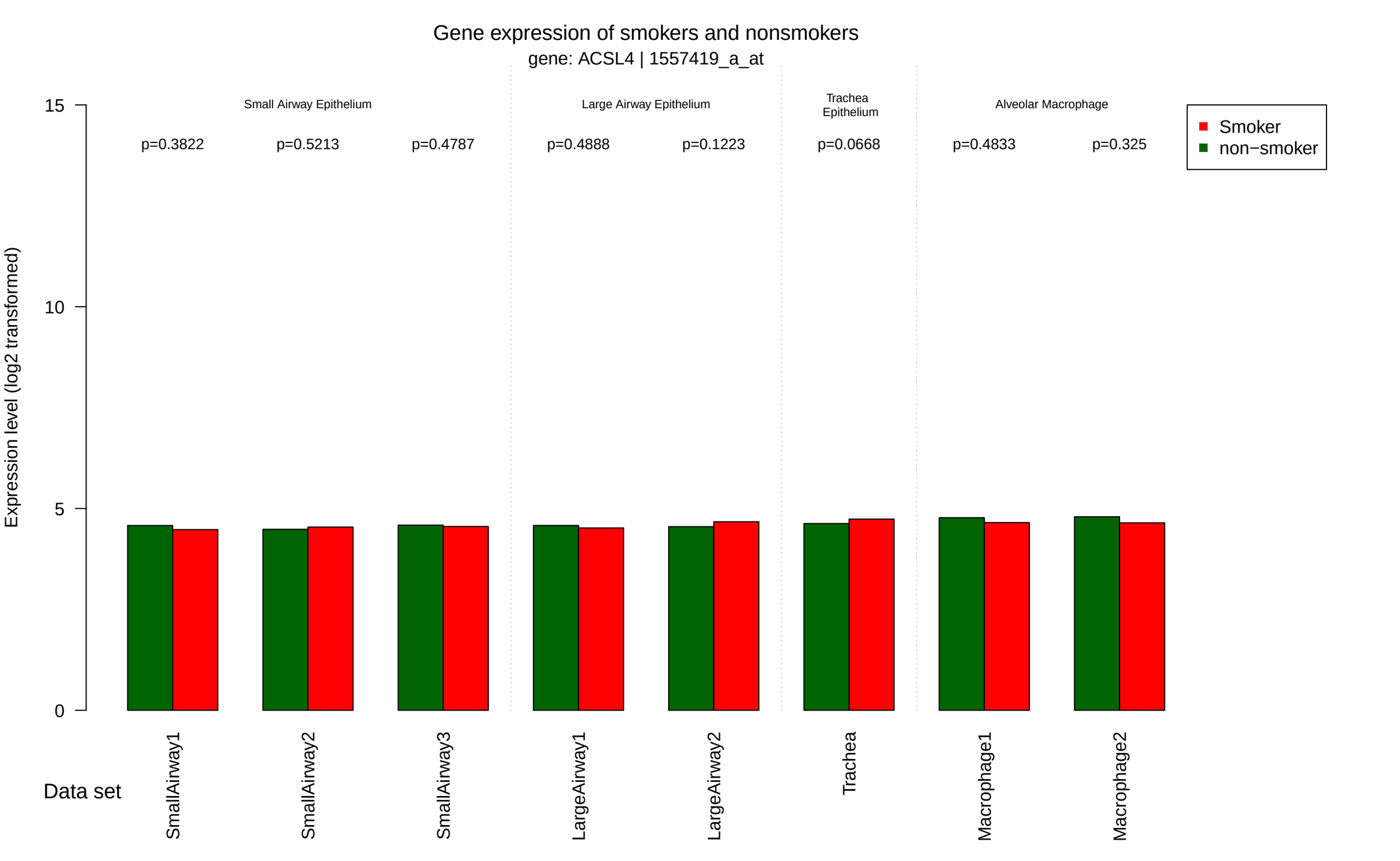
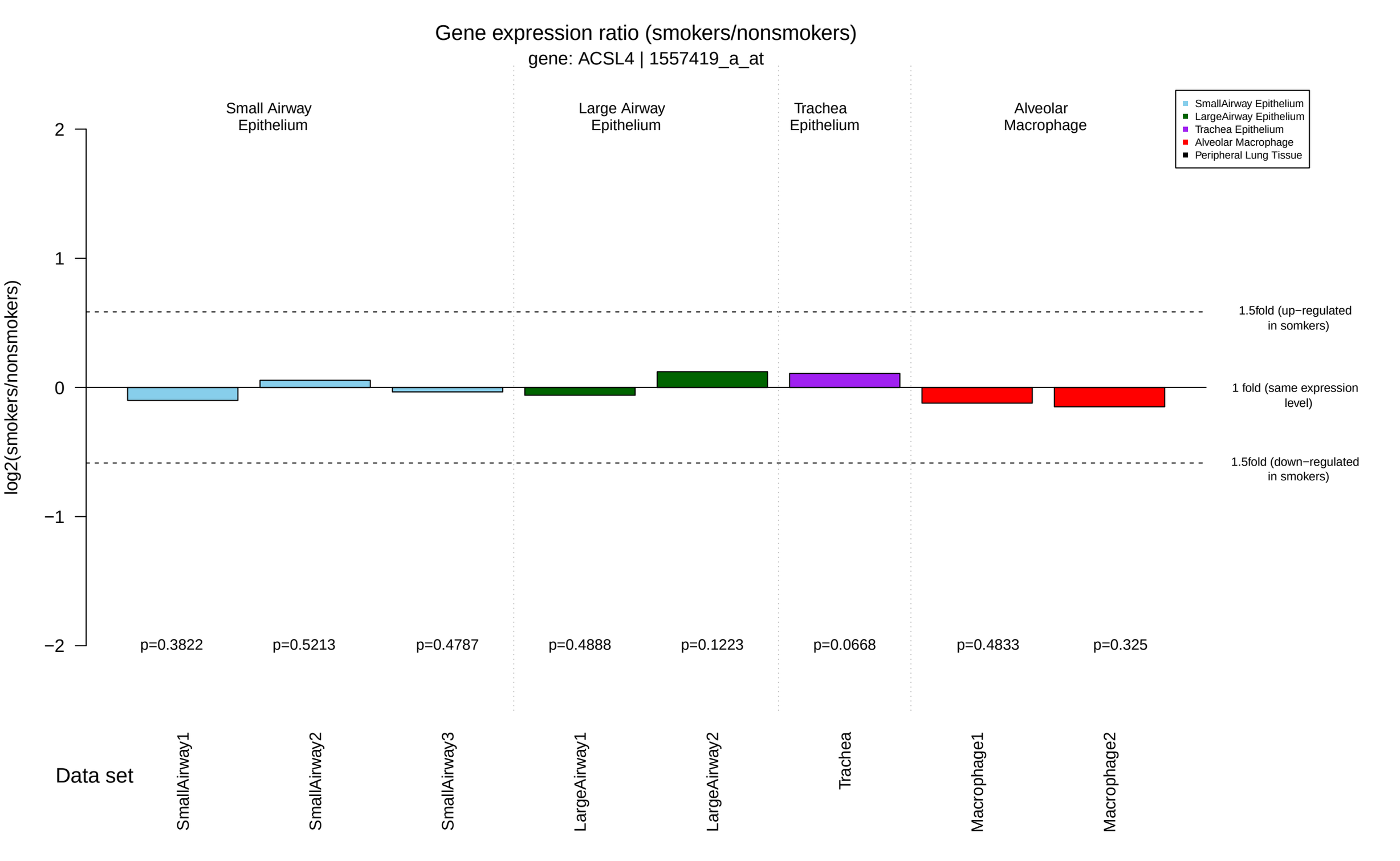
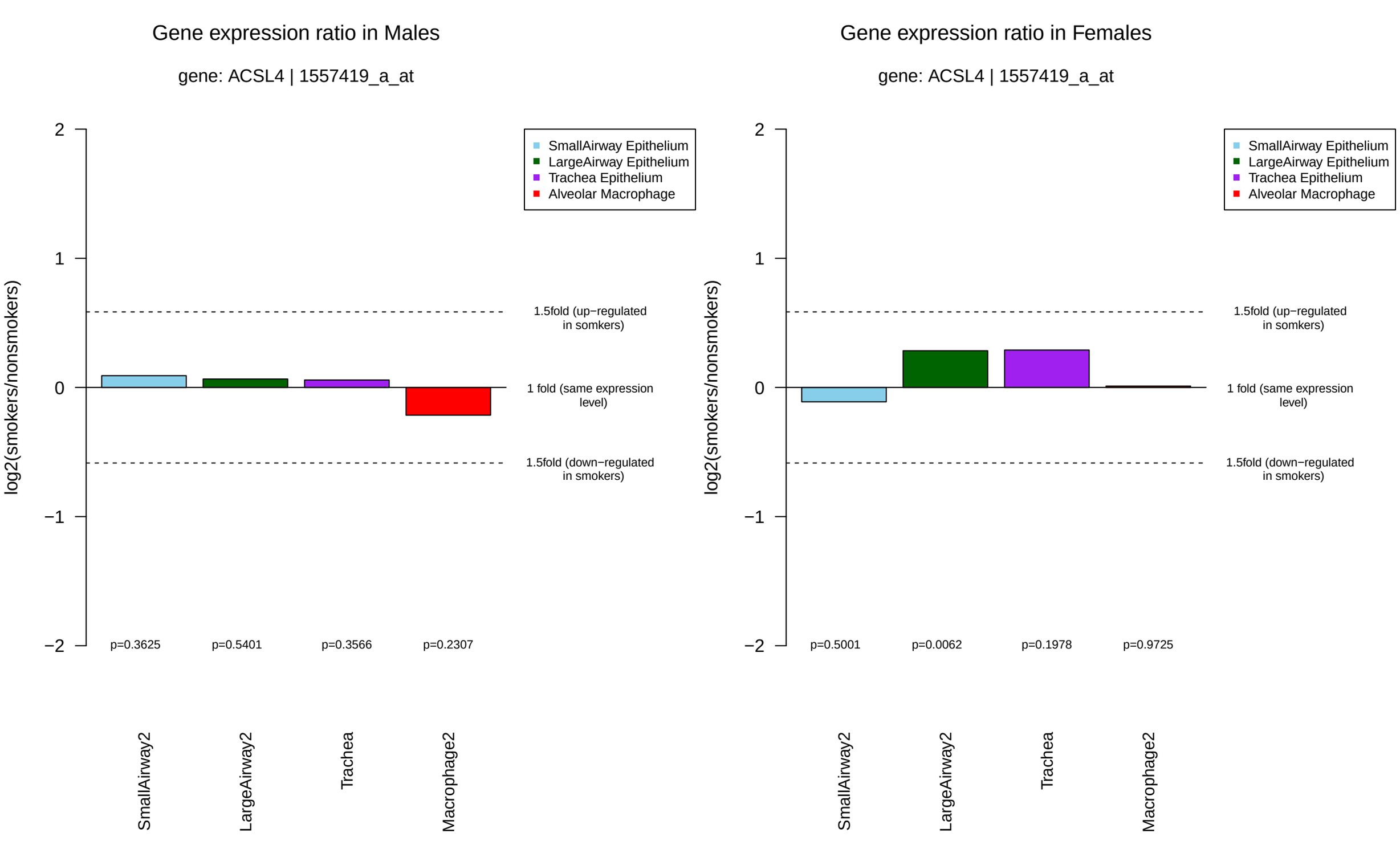
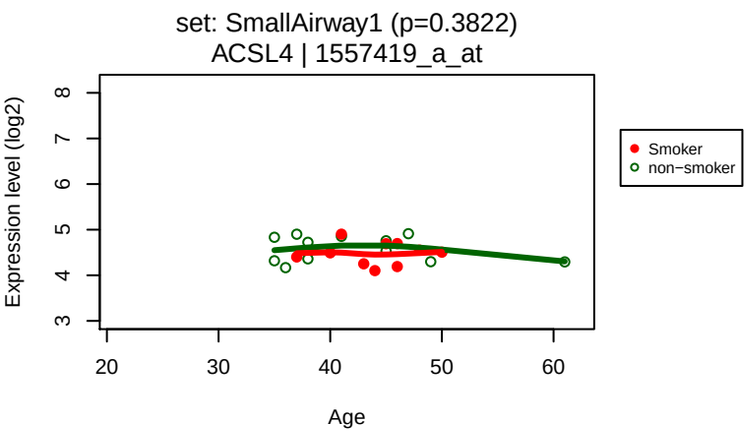
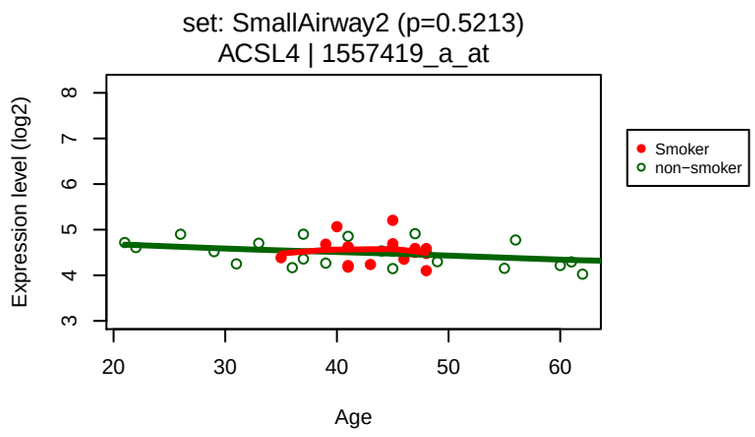
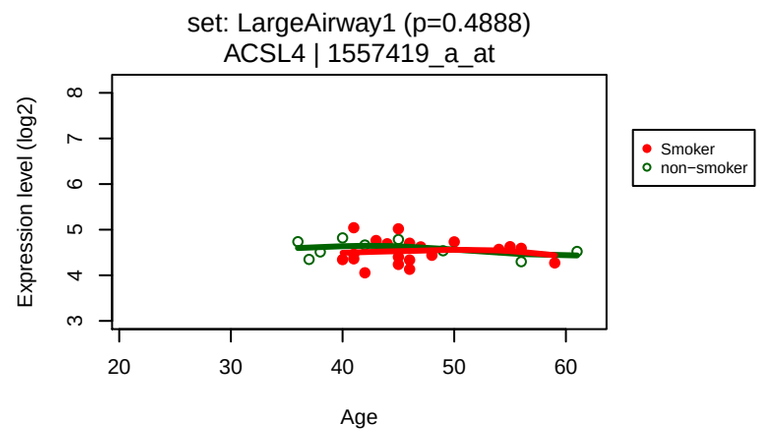

Gene expression of smokers and nonsmokers
gene: ACSL4 | 1557419_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3822
p=0.5213
p=0.4787
p=0.4888
p=0.1223
p=0.0668
p=0.4833
p=0.325
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

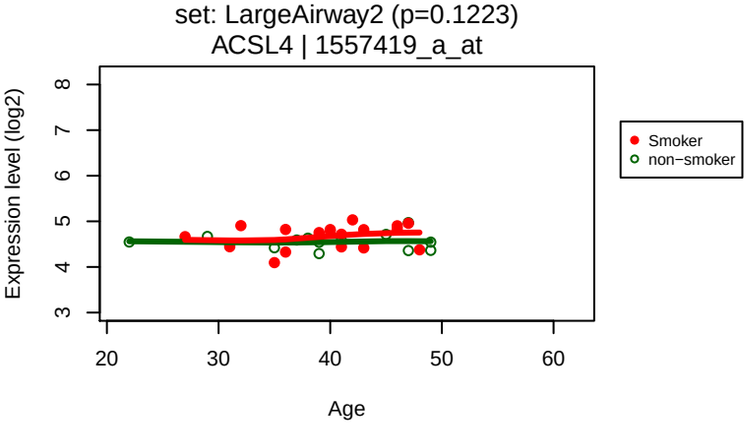

Gene expression ratio (smokers/nonsmokers)
gene: ACSL4 | 1557419_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3822
p=0.5213
p=0.4787
p=0.4888
p=0.1223
p=0.0668
p=0.4833
p=0.325
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

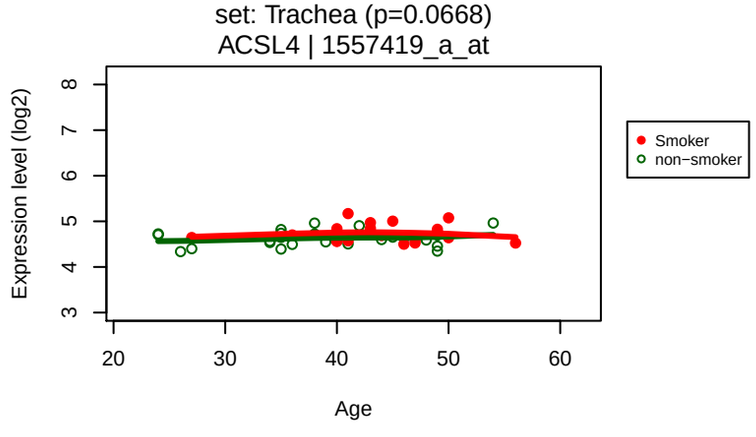

Gene expression ratio in Males
gene: ACSL4 | 1557419_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3625
p=0.5401
p=0.3566
p=0.2307
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ACSL4 | 1557419_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5001
p=0.0062
p=0.1978
p=0.9725
SmallAirway2
LargeAirway2
Trachea
Macrophage2

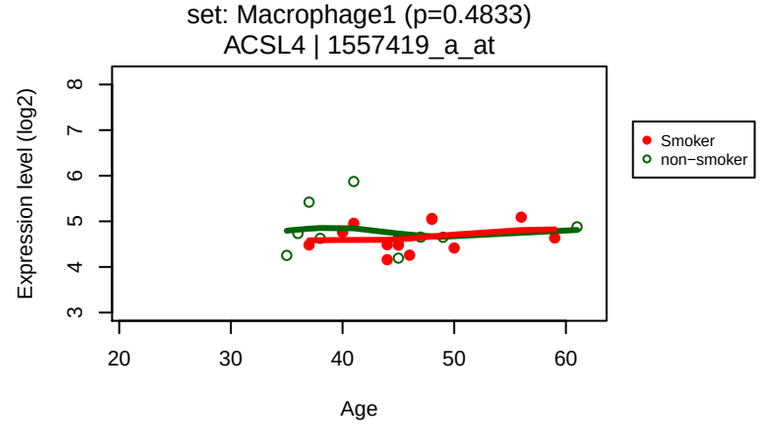

set: SmallAirway1 (p=0.3822)
ACSL4 | 1557419_a_at
Expression level (log2)
Age
Smoker
non−smoker

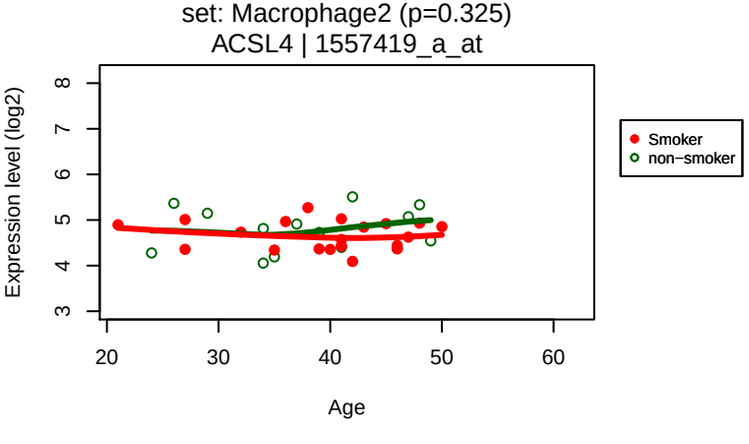

set: SmallAirway2 (p=0.5213)
ACSL4 | 1557419_a_at
Expression level (log2)
Age
Smoker
non−smoker

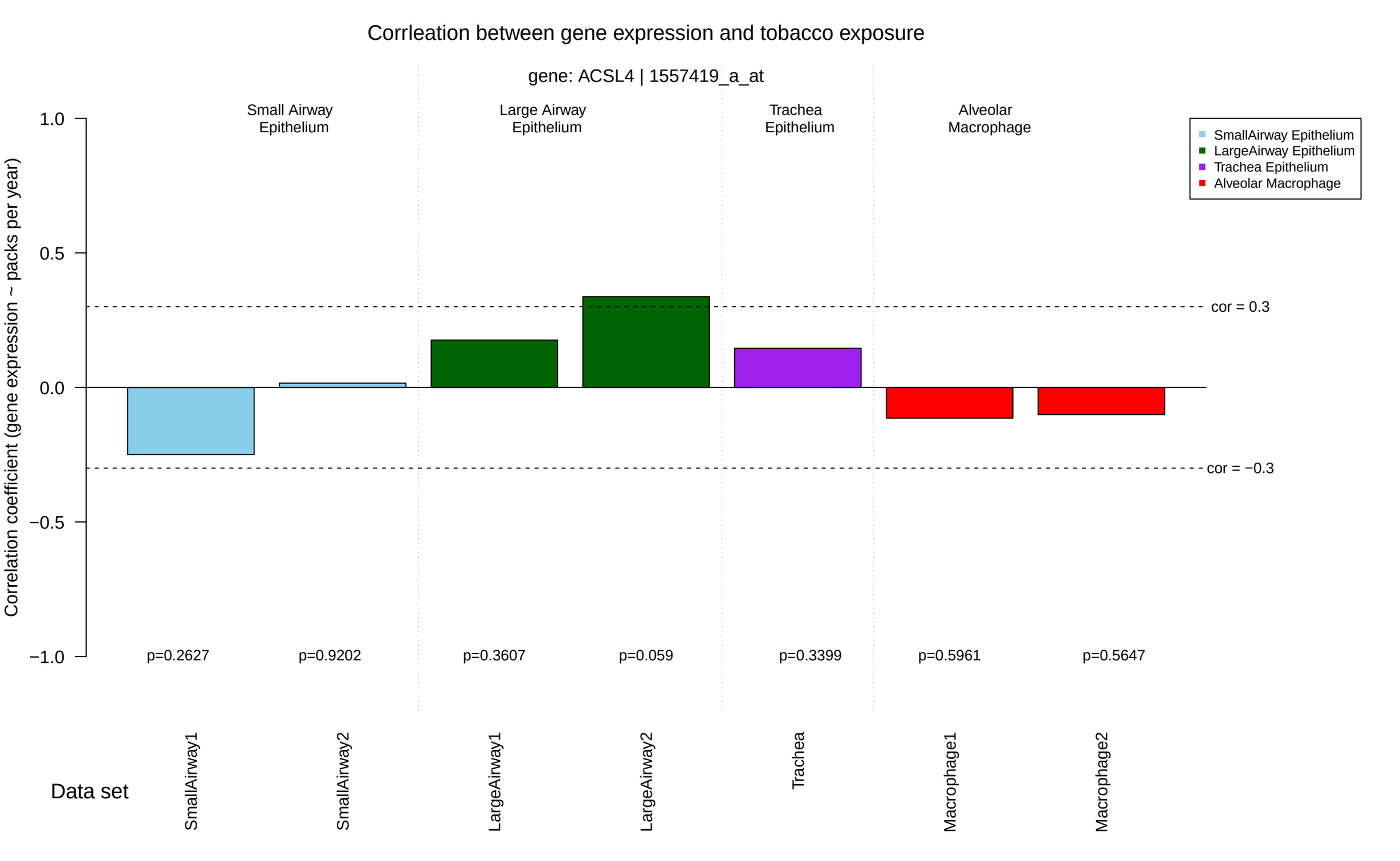

set: LargeAirway1 (p=0.4888)
ACSL4 | 1557419_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1223)
ACSL4 | 1557419_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0668)
ACSL4 | 1557419_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4833)
ACSL4 | 1557419_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.325)
ACSL4 | 1557419_a_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ACSL4 | 1557419_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2627
p=0.9202
p=0.3607
p=0.059
p=0.3399
p=0.5961
p=0.5647
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2